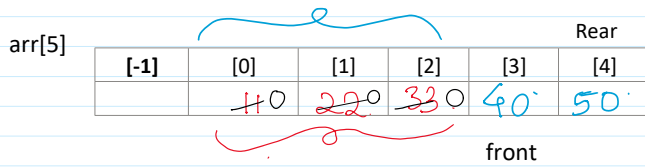


Linear Queue : Linear Data structure : utility DS

06 May 2025 13:57



Empty ()
 if (rear == -1 || front > rear)

Front end : pop (dequeue)
 Rear end : push (enqueue)

Push (enqueue) :

- 1) Increment the rear by 1
- 2) Insert the element at the rear position
- 3) If (front == -1) front = 0;

Pop (dequeue) :

- 1) Arr[front] = 0 (optional)
- 2) Increment the front by 1

Queue full condition : rear == size-1

Queue empty condition : rear == -1 || front > rear

Always check if the queue is full before inserting the element

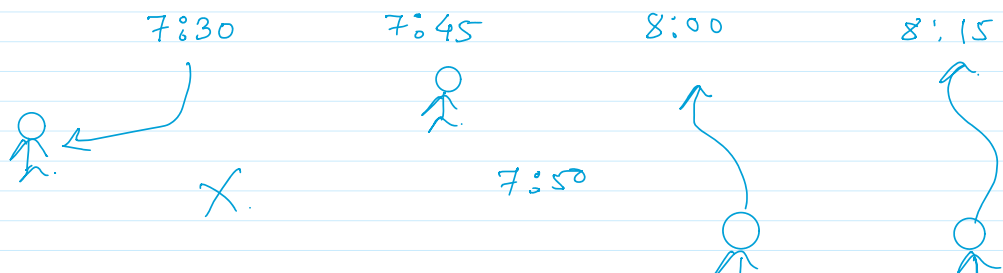
Always check if the queue is empty before deleting the element

push() / Enqueue() / insert()
 {
 rear++;
 arr[rear] = value;
 if (front == -1)
 front = 0;
 }

int peek()
 return arr[front];

pop() / dequeue() / delete()
 {
 arr[front] = 0;
 front++;
 }

7:30 to 8:30



X.

Toss

